

ADAS with Real Time Interactive Display

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Introduction & Motivation

Introduction

- Advanced Driver Assistance Systems (ADAS) enhances vehicle safety and driving experience by using sensors, cameras, and software to monitor surroundings and take control over the car if necessary.
- ADAS enhances safety for both driver and passengers by utilizing advanced technologies such as Artificial Intelligence and Machine Learning.
- These systems employ sensors and high-performance computing units to monitor the vehicle's surroundings and assist the driver in real-time and display the information on the dashboard, thereby significantly reducing the risk of accidents caused by human negligence.

Motivation

The primary motivation for selecting an ADAS based on low-end computing as our project topic stems from the increasing number of road accidents.

Problem Statement

Problem Statement

In urban and highway driving environments, human error remains a leading cause of traffic accidents. There is a critical need for an affordable, real-time driver assistance solution capable of detecting road signs, interpreting traffic light signals, and identifying lane boundaries accurately. Existing commercial ADAS systems are either expensive or lack adaptability for custom embedded platforms like Raspberry Pi.

Objectives of the Project

1. **Lane Keep Assist:** Implement lane-keeping and lane departure warning features to enhance lane discipline.
 2. **Traffic Sign and Signal Detection:** Develop systems to detect and respond to traffic signs and signals.
 3. **Collision Avoidance:** Integrate obstacle detection and avoidance mechanisms to prevent collisions.
 4. **Real-Time Interactive Display:** Developing a dashboard with an LCD screen to showcase traffic signs, signals, and other outputs, to enhance driver understanding.
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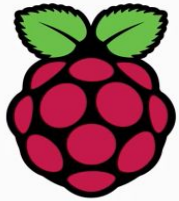
Literature Review

Research papers	Descriptions	Results
Lane Detection Techniques using Image Processing [1]	Lane detection is essential for autonomous navigation, but most systems struggle with unmarked or poorly marked roads. This study reviews a lane detection approach for such conditions, proposing an improved method using digital image processing and camera data. The goal is to provide real-time curve values, aiding drivers or autonomous vehicles in safely navigating roads with unclear or absent markings.	The study presents an improved lane detection approach for unmarked roads using digital image processing and camera data. The system provides real-time curve values, helping drivers or autonomous vehicles navigate turns safely and prevent accidents, even without clear road markings, thereby enhancing overall road safety.
Real-time Lane Detection Method Based On Region Of Interest [2]	Video Frame Preprocessing: Enhance image contrast and apply an advanced multi-angle edge detection algorithm to suppress non-lane edges while preserving lane features. Region of Interest Model: Use the Hough Transform to identify straight lane segments and fit dotted lines with the least squares method. Vanishing Point Calculation: Calculate the intersection of lane lines to predict direction based on the vanishing point's position relative to the image center.	● Video Frame Preprocessing: Enhanced image contrast through nonlinear transformation and applied an improved multi-angle edge detection algorithm to suppress non-lane edges while preserving lane features. ● Region of Interest Model: Improved the Hough Transform algorithm to identify straight lane segments and fit dotted lane lines using the least squares method, extracting continuous lane lines.

Literature Review

Research papers	Descriptions	Results
Traffic Sign Detection and Recognition using a CNN Ensemble [3]	Drivers often miss roadside signs while focusing on the road, creating safety risks. Traffic Sign Detection and Recognition technology can address this by automatically detecting and identifying signs, alerting drivers without requiring them to look away. This improves driver awareness and road safety by reducing missed signs while keeping attention on the road.	The paper concludes that Traffic Sign Detection and Recognition effectively alerts drivers to upcoming signs without diverting their attention. The model uses Stochastic Gradient Descent with Nesterov momentum and a categorical cross-entropy loss function, achieving an accuracy of 98.11%.
Traffic sign detection and recognition based on convolutional neural networks [4]	This paper presents a Traffic Sign Detection and Recognition system that classifies 28 globally recognized traffic signs, unlike studies focusing on a limited set. It uses two neural networks: one for sign classification and another for shape detection. Trained on 40,000 images with data augmentation, the first classifier uses 28,000 positive and 12,000 negative images, while the second uses 2,400 positive and 1,200 negative images. The system processes images to identify the region of interest, which is then classified by two CNNs.	The proposed system effectively detects and classifies 28 globally recognized traffic signs using two separate CNN classifiers for sign and shape detection. Trained on 40,000 images, it achieved high accuracy. Image processing for region identification improved performance, showcasing the system's effectiveness in real-world traffic sign recognition.

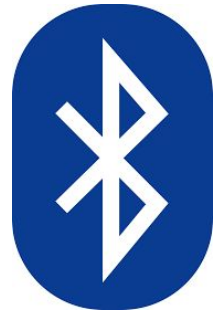
Software Requirements



Raspberry Pi OS



TensorFlow Lite



Arduino IDE

Hardware Requirements



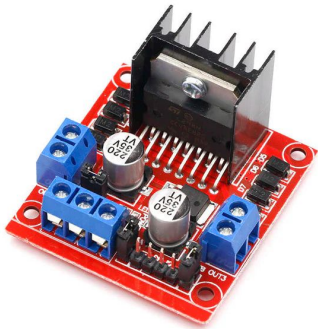
Vehicle Chassis



XL6019



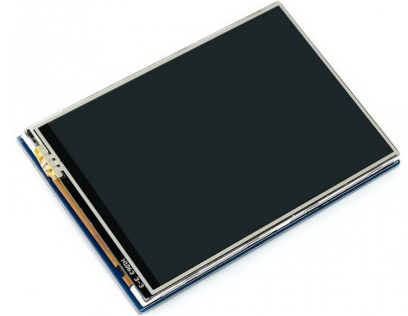
Arduino Mega



L298N Motor Driver



Raspberry Pi Model 4

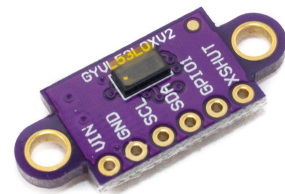


LCD Display

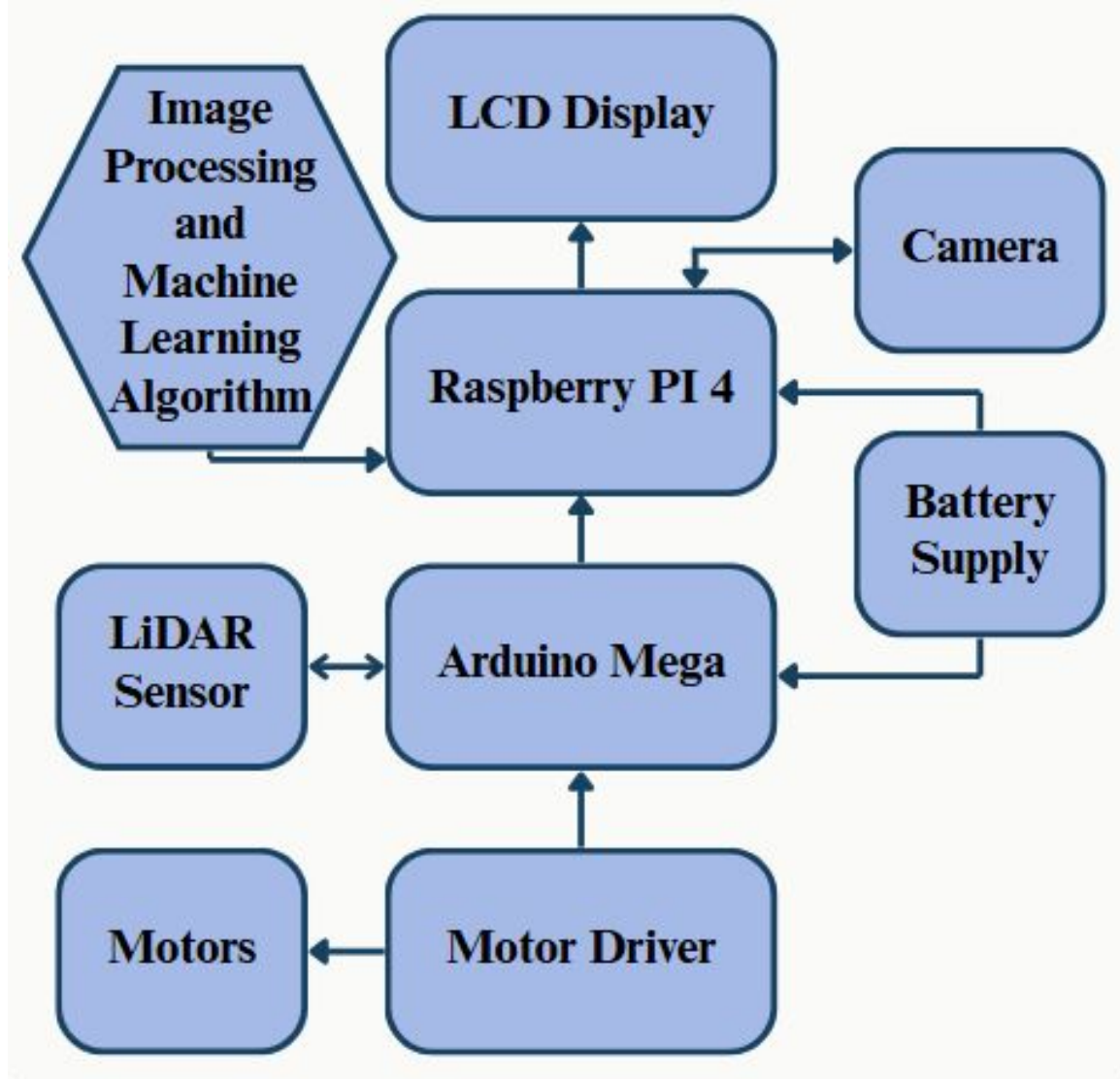


HC-05

LiDAR
Sensor



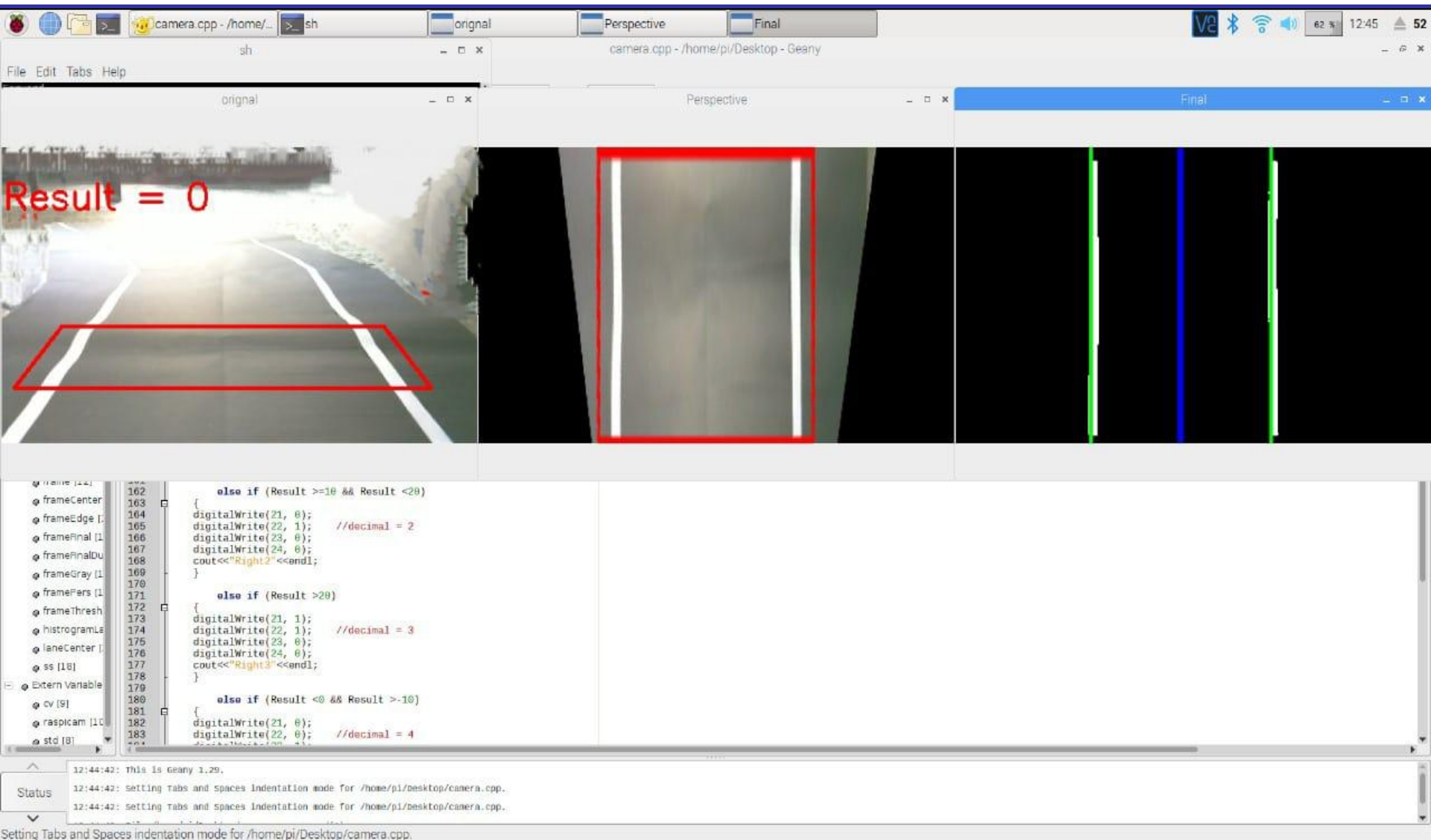
Flowchart



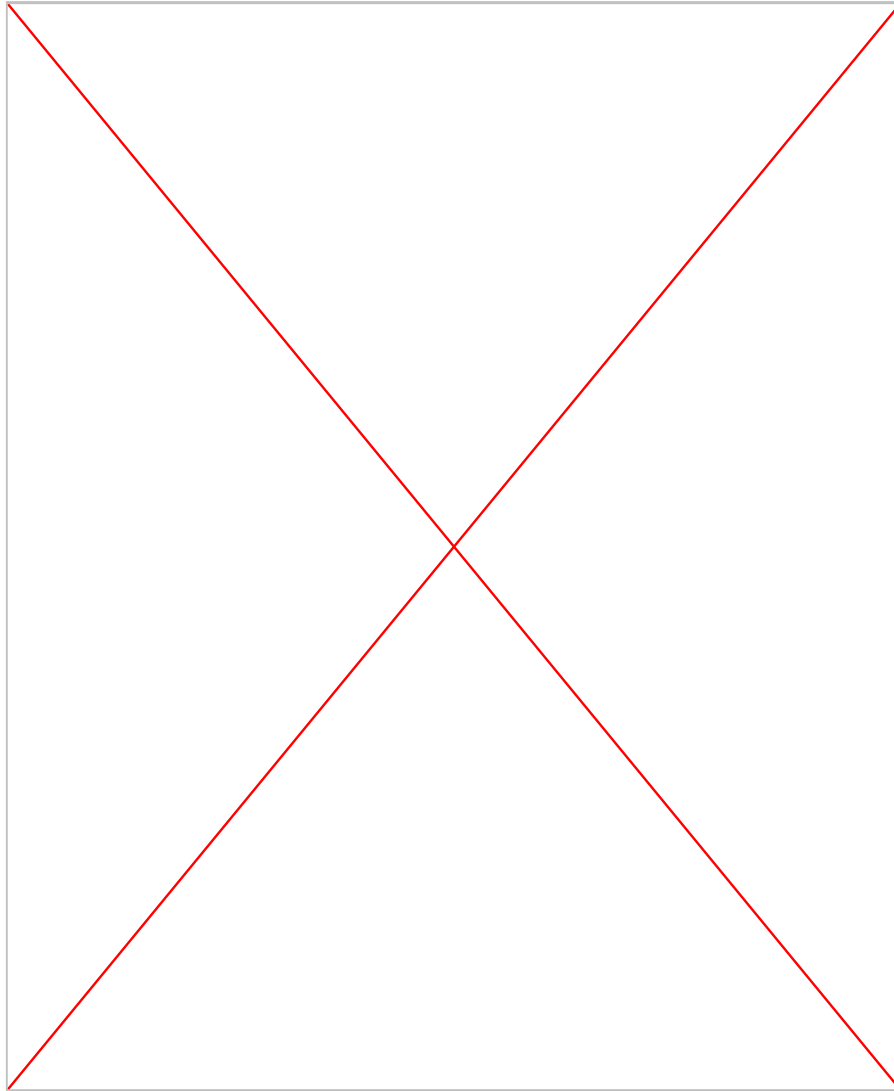
Desired model



Lane Detection



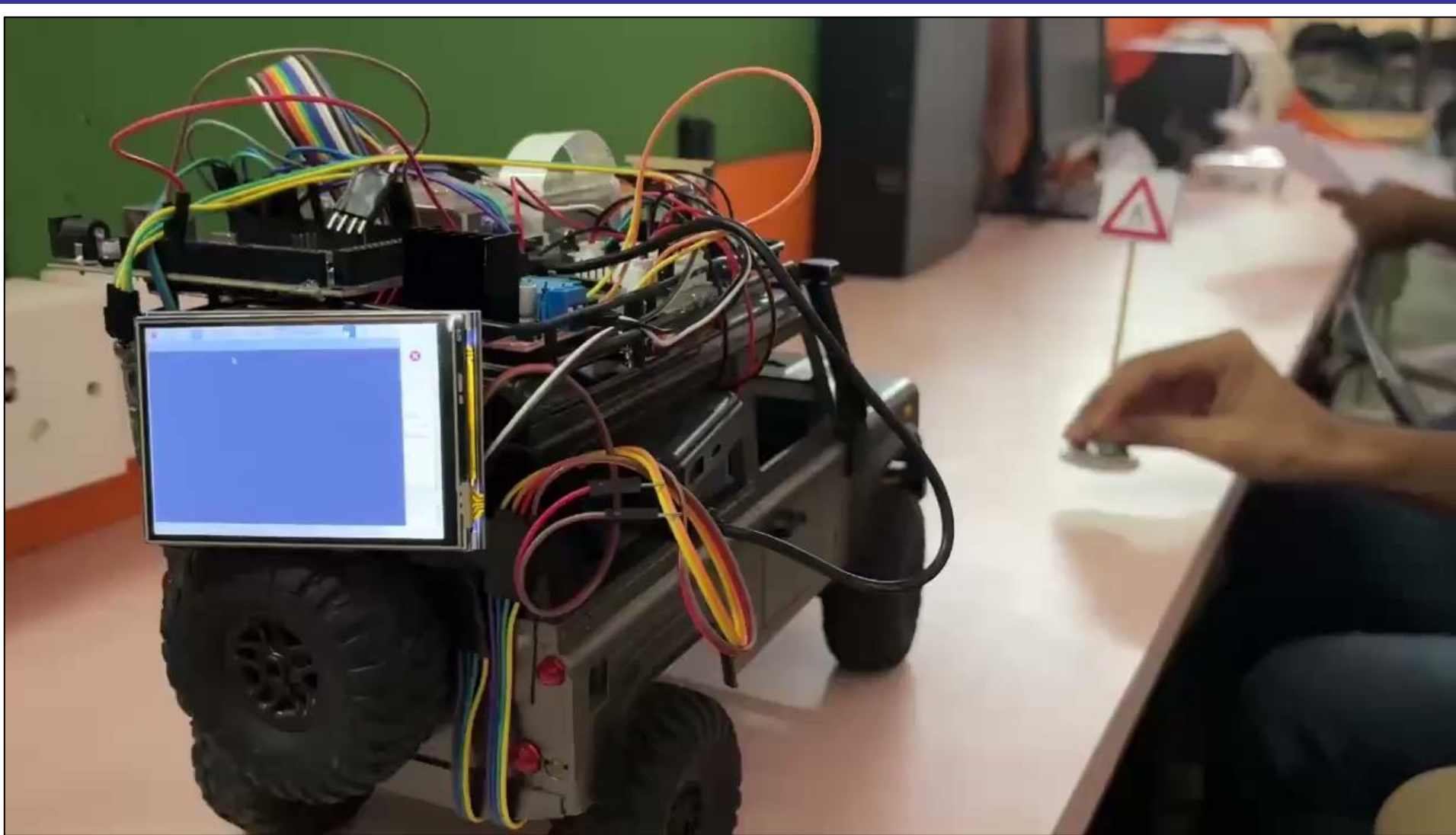
Lane Keep Assist



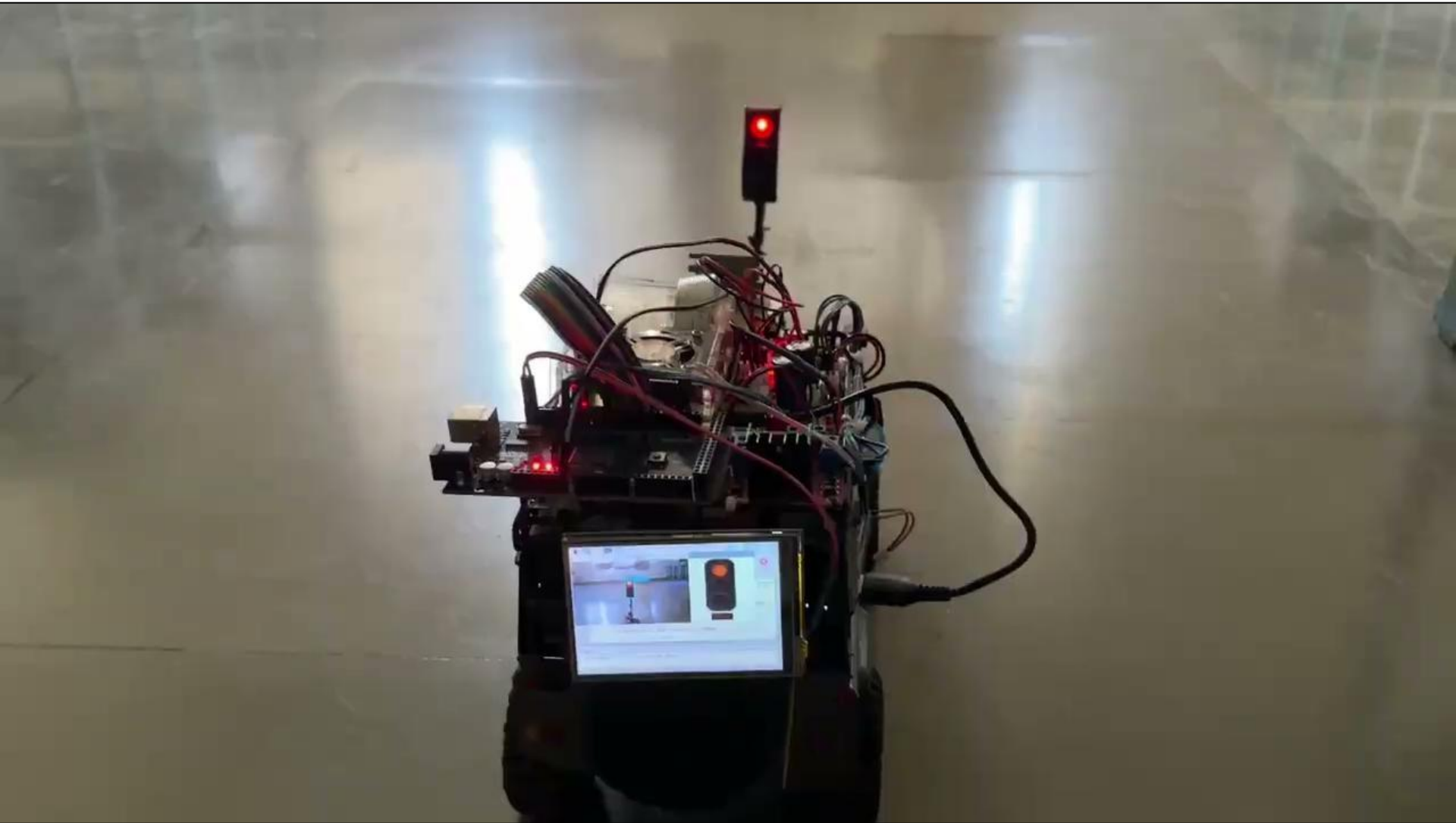
Traffic Sign Detection



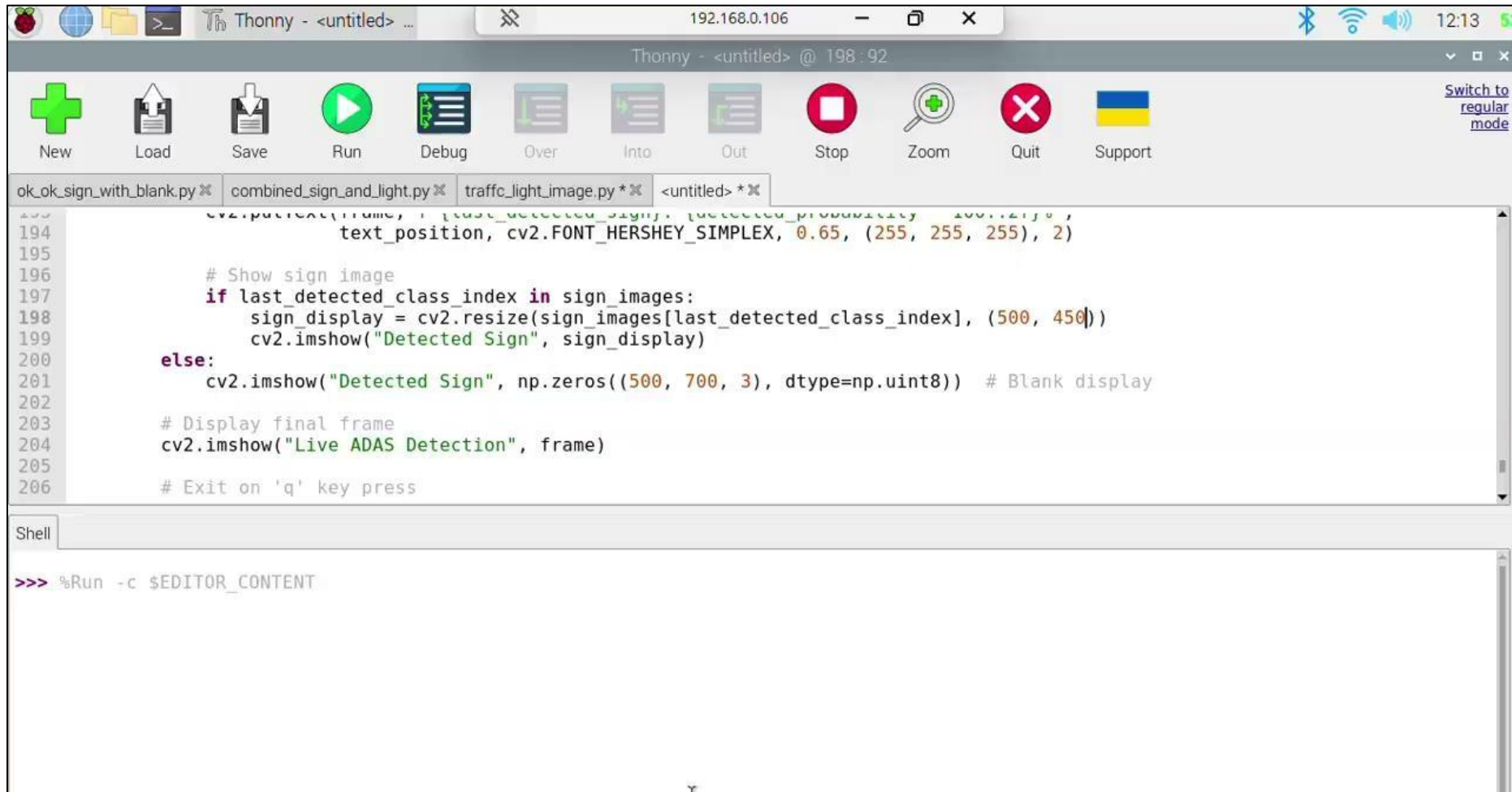
Traffic Sign Live Detection



Live Traffic Signal Detection



Traffic light and Sign Detection

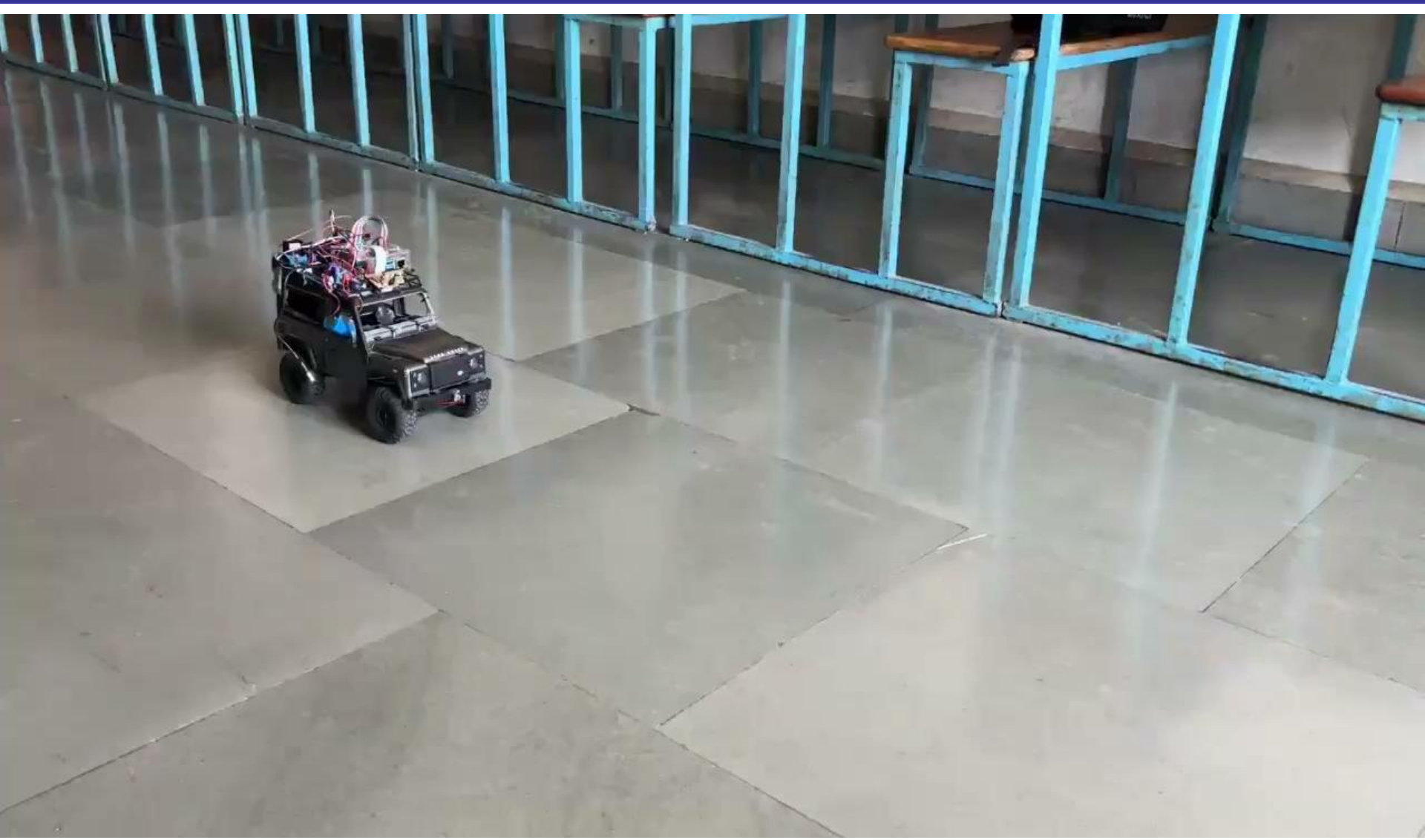


The screenshot shows the Thonny Python IDE interface. The top bar displays the IP address 192.168.0.106 and the time 12:13. The toolbar includes icons for New, Load, Save, Run, Debug, Over, Into, Out, Stop, Zoom, Quit, and Support. The file explorer shows three open files: ok_ok_sign_with_blank.py, combined_sign_and_light.py, and traffic_light_image.py. The main editor window displays a Python script for traffic light and sign detection. The script includes comments and code for displaying the final frame and exiting on 'q' key press. The shell window at the bottom shows the command >>> %Run -c \$EDITOR_CONTENT.

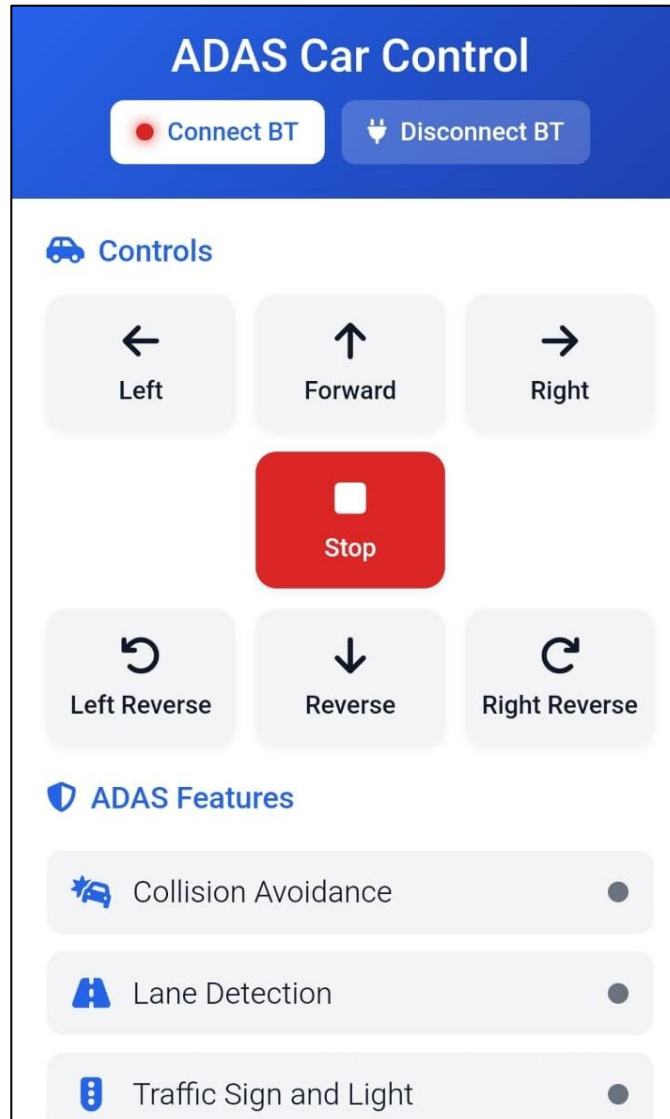
```
194         cv2.putText(frame, f'last_detected_sign: {detected_probability}',
195                       text_position, cv2.FONT_HERSHEY_SIMPLEX, 0.65, (255, 255, 255), 2)
196
197     # Show sign image
198     if last_detected_class_index in sign_images:
199         sign_display = cv2.resize(sign_images[last_detected_class_index], (500, 450))
200         cv2.imshow("Detected Sign", sign_display)
201     else:
202         cv2.imshow("Detected Sign", np.zeros((500, 700, 3), dtype=np.uint8)) # Blank display
203
204     # Display final frame
205     cv2.imshow("Live ADAS Detection", frame)
206
207     # Exit on 'q' key press
```

```
>>> %Run -c $EDITOR_CONTENT
```


Collision Avoidance



Android Studio App UI



Conclusion

- This project aims to develop an ADAS using Raspberry Pi 4, Arduino Mega, Electric motors, LCD display, and a Lidar sensor.
- The system focuses on improving road safety by detecting traffic signals and signs, providing lane keep assist, and collision avoidance.
- By integrating machine learning, we implemented the real time traffic sign detection system. and also to display the signs on the LCD screen
- This project not only advances automotive safety technologies but also offers hands-on experience in modern software development and system integration.

References

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2. Y. Zhang, P. Gong, S. Ji and Q. Xu, "Real-time Lane Detection Method Based On Region Of Interest," 2022 International Conference on Intelligent Transportation, Big Data & Smart City (ICITBS), Hengyang, China, 2022, pp. 1188-1192, doi: 10.1109/ICITBS55627.2022.00255. keywords: {Visualization;Lane detection;Image edge detection;Roads;Lighting;Feature extraction;Prediction algorithms;Lane detection;Nonlinear transformation;Multi-angle edge detection;Region of Interest}
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4. Y. Sun, P. Ge and D. Liu, "Traffic Sign Detection and Recognition Based on Convolutional Neural Network," 2019 Chinese Automation Congress (CAC), Hangzhou, China, 2019, pp. 2851-2854, doi: 10.1109/CAC48633.2019.8997240. keywords: {Image color analysis;Filtering;Transforms;Training;Image segmentation;Roads;Feature extraction;traffic sign recognition;traffic sign detection;deep learning;convolutional neural network},